SANGER: CLONED AMPLICONS

one clone is one molecule, read once

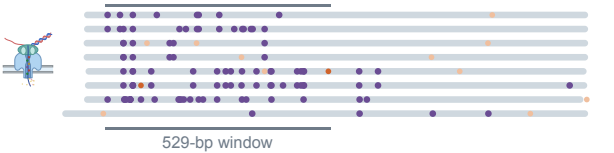


529-bp window that the clones read cleanly;
447 chromatograms from the six mice of Figure 5

● mutation ● sequencing error

NANOPORE: CLONING-FREE AMPLICONS

every read spans the 1,230-bp amplicon



Eight wild-type reads over the whole amplicon, with every mutation called in them; errors drawn at the rate the controls show, 1 base in 700, pale below Q 20

ILLUMINA: PAIRED SHORT READS

each pair is merged into one read, which then takes the same read route as nanopore



J_H4 (Ldha study): 143-bp fragment, 2 × 151-bp reads

R1 → adapter, dropped
← R2

merged

every base read twice; 95% of pairs merge

CH12F3 VDJ: 269-bp fragment, 2 × 150-bp reads

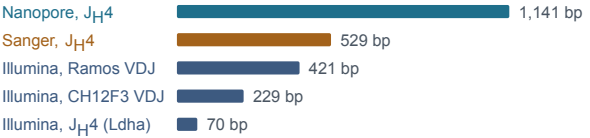
R1 →
← R2

merged

31 bp read twice, the rest once; 97% merge

read twice (Q60) read once | disagreement: better call kept, N on a tie

How much of the locus one read covers



Illumina settings: minimap2 sr preset; error gate 0.02 per base; minimum alignment ≤ 100 bp; then the same control test and FDR. The Ldha and CH12F3 files carry a flat Q30 (SRA), so their error model has a single quality level; the Ramos files keep binned ones.

From the Sanger clones here

Each molecule's own mutations
the mutation load of every clone, the share of clones mutated, and frequency as published: unique mutations over clones × a fixed length

Bookkeeping checks
copied traces, clones filed under two mice, mutated clones filed as controls, mislabeled tubes (Figure S7)

Error reported, not subtracted
the AID-null clones' error rate (about 20 per 10^5 bases at Q ≥ 20 in both published studies), and how many of each animal's calls it explains

From both, and they agree

Wild type against AID-null
Sanger 38-fold; nanopore 141-fold, with controls without calls counted at one event

Substitution spectrum
r = 0.92 (95% CI 0.74–0.98), slope 0.97

Position along the window
r = 0.94 in 25-bp bins

AID hotspot use
G:C at WRC/GYW 58% and 54% (20% of sites); A:T at WA/TW 70% and 69% (50% of sites)

Not comparable: absolute density
a clone is read once; nanopore counts unique mutations among 500 reads. Compare densities within a platform, not between.

From the nanopore reads here

The whole amplicon
positions 90–1,230: 229-fold over controls; 69% of mutations in the first 450 bp, falling 7-fold to the last 250 bp (Figure 6A)

Clones and mutation trees
clones of each mouse, and their mutations ordered into trees (Figures 6 and S11)

Calls at an estimated false-discovery rate
each candidate tested against the controls, the threshold set by them as decoys; single-read mutations not called, as one background-subtracted estimate